

VAISHNAVI SINGH

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PROFESSIONAL SUMMARY

Ambitious and results-oriented Computer Science Engineering graduate with a stellar academic record (CGPA 8.50/10) and a demonstrated aptitude for Java development, object-oriented architecture, and backend engineering. Adept at translating theoretical knowledge into functional, scalable applications through hands-on project development. Proven ability to independently architect and deploy end-to-end solutions using Java, Spring Boot, MySQL, and React.js. Thrives in collaborative, agile environments and is passionately committed to continuous learning, clean code practices, and delivering high-impact software solutions.

TECHNICAL SKILLS

Programming Languages: Java (Proficient), C, C++, JavaScript (ES6+), ReactJS

Frameworks & Libraries: Spring Boot, Hibernate ORM, React.js, REST API Design

Databases: MySQL (Relational Queries, Joins, Indexing), MongoDB (NoSQL, Document Modeling)

Engineering Concepts: Object-Oriented Programming, MVC Architecture, CRUD Operations, Data Structures, Debugging & Optimization

Tools & Platforms: Git, GitHub, IntelliJ IDEA, VS Code, Postman, Eclipse

Soft Skills: Analytical Problem-Solving, Agile Adaptability, Technical Documentation, Remote Collaboration

PROJECTS & REAL-WORLD EXPERIENCE

E-Commerce REST API Backend

2024 | Java · Spring Boot · MySQL · Postman

- Architected and implemented a fully functional RESTful API for an e-commerce platform, enabling seamless product listing, cart management, and order processing workflows.
- Engineered secure user authentication and role-based authorization leveraging Spring Boot's security framework, safeguarding sensitive transactional data.
- Designed a normalized relational database schema in MySQL (10+ tables) with optimized JOIN queries, reducing data retrieval latency by an estimated 35%.
- Validated all API endpoints using Postman, ensuring 100% functional compliance with defined business logic before integration.
- Applied MVC architectural pattern and adhered to SOLID principles, resulting in a maintainable, modular codebase ready for horizontal scaling.

Student Management System

2024 | Java · JDBC · MySQL · Java Swing

- Conceived and delivered a comprehensive desktop application to automate student record management including enrollment, attendance tracking, grade computation, and report generation.
- Established a robust Java JDBC pipeline to interface seamlessly with a MySQL database, executing parameterized queries to prevent SQL injection vulnerabilities.
- Designed an intuitive Java Swing GUI, reducing administrative data-entry time by approximately 40% through streamlined form-based interactions.
- Incorporated dynamic search and filter functionality enabling instantaneous retrieval of student profiles across thousands of records.
- Implemented exception-handling mechanisms and data validation layers to ensure system resilience and data integrity under varied edge-case inputs.

Task Management Web Application

2025 | React.js · JavaScript · HTML5 · CSS3

- Developed a responsive, single-page task management application using React.js with component-driven architecture, enabling users to create, prioritize, and track daily deliverables.
- Leveraged React Hooks (useState, useEffect) to orchestrate real-time UI state updates, delivering a fluid and interactive user experience without page reloads.
- Integrated localStorage for client-side data persistence, ensuring task continuity across browser sessions without requiring a backend infrastructure.
- Applied responsive CSS Grid and Flexbox layouts, achieving seamless cross-device compatibility across mobile, tablet, and desktop viewports.
- Implemented drag-and-drop task reordering and color-coded priority tagging to enhance user productivity and visual clarity.

Java OOP Mini-Projects Suite

2023–2024 | Core Java · OOP · Collections Framework

- Engineered a diverse suite of Java programs—including a Library Management System, Banking Simulator, and Student Grade Calculator—to systematically master object-oriented design principles.
- Leveraged Java Collections Framework (ArrayList, HashMap, LinkedList) to design efficient data structures tailored to each application's computational requirements.
- Practiced encapsulation, inheritance, polymorphism, and abstraction across multi-class hierarchies, reinforcing industry-grade coding conventions.
- Performed systematic debugging and code refactoring cycles, achieving cleaner, DRY-compliant codebases with measurably improved runtime efficiency.

EDUCATION

B.E. in Computer Science & Engineering

Dec 2021 – Jan 2026

Nitte Meenakshi Institute of Technology, Bengaluru | CGPA: 8.50 / 10

Pre-University Certificate (PUC)

March 2023 – March 2024

Army Public School, K. Kamraj Road | Score: 564 / 600 (94%)

CBSE Class X Board Examination

July 2020 – July 2021

Army Public School, MG Road | Percentile: 95%

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Java Programming Certification – Infosys Springboard (October 2025)
- Object-Oriented Programming Using Java – Completed with Distinction
- MongoDB Developer Certification – NoSQL Data Modeling & Aggregation Pipelines
- Backend Development Fundamentals – RESTful Services & HTTP Protocol
- Programming Foundations – Algorithms, Data Structures & Complexity Analysis

ACHIEVEMENTS & RECOGNITION

- Secured 1st Place in Code Quest inter-collegiate competitive programming event, demonstrating exceptional algorithmic problem-solving under time constraints.
- Cleared Smart India Hackathon (SIH) Round 1 (October 2025) – selected among top teams to propose innovative technology-driven solutions to national challenges.
- Awarded Best Theme Award at the National Robotics Championship (2019) – recognized for outstanding creativity and engineering innovation at a national level.

ADDITIONAL INFORMATION

- Actively available for a full-time remote internship engagement (1–3 months); immediate joiner.

- Consistently demonstrates rapid skill acquisition, having independently mastered Spring Boot and MongoDB through self-directed study alongside formal coursework.
- Exhibits strong remote-work discipline, time-management proficiency, and a proactive approach to collaborative team communication.